Consistent Hashing → sharding

Redis → Buckets → 1 Bucket per person

Redis Instance / server → 200K

1B → 10% of 1B → 100M → Redis

R1 R2 R3 → horizontal scaling ?

X L.B → NO → sharding → 1→100K R1 101 → 200K R2

150K

S6
(6) → { S1 S5 { S2 } S3 S4
% 6

100 servers

userid ⇒ h(uid) ⇒ 1067 % 100 → 0 → 99 (67)

↳ 67th server

Problem

(4) → (3)

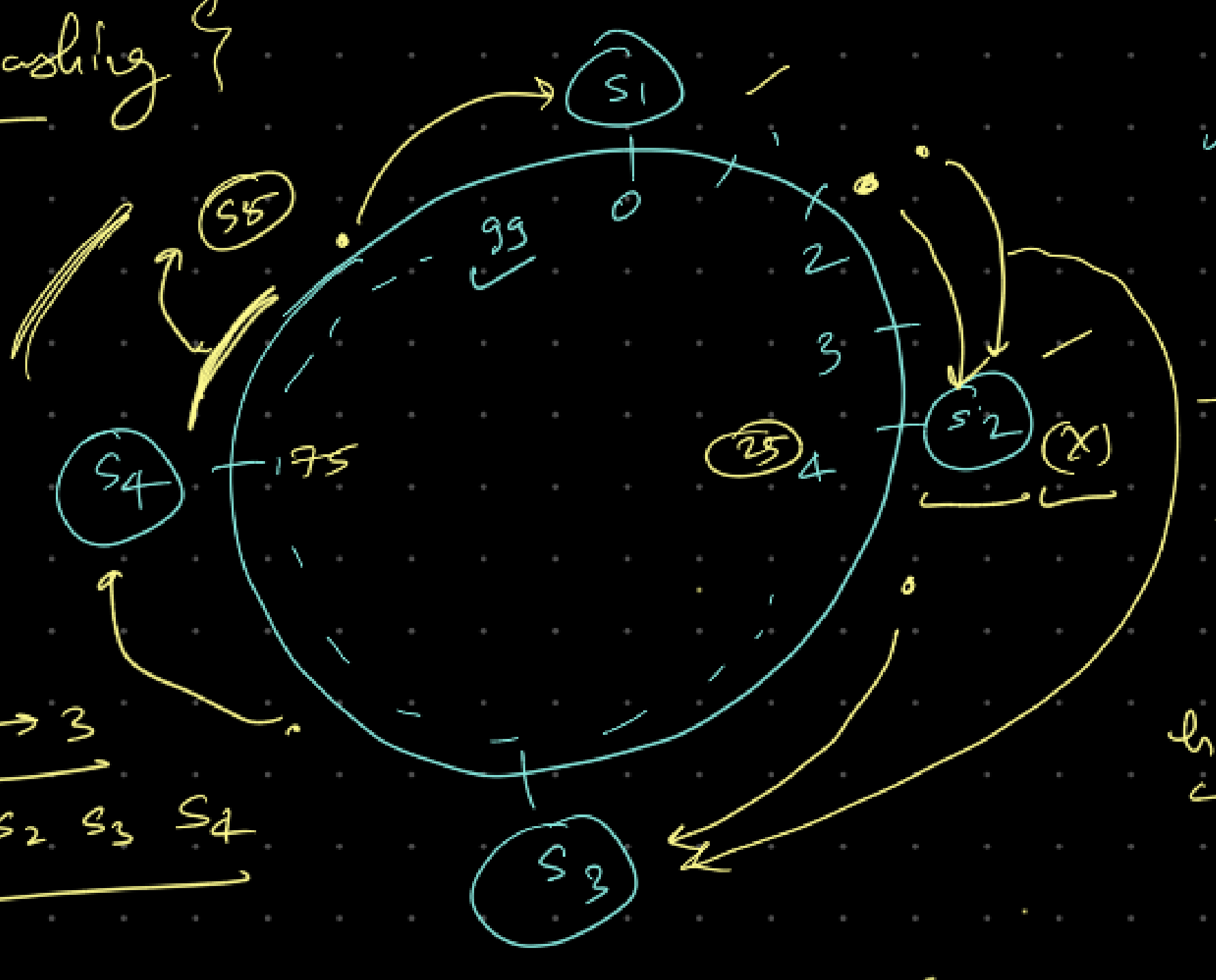
% 4 → % 3

4 99%

Removing / Adding ⇒ {Traffic Redistributes}

Goal → {minimize this redistribution} ✓

{Consistent Hashing}



4 servers

100 slots

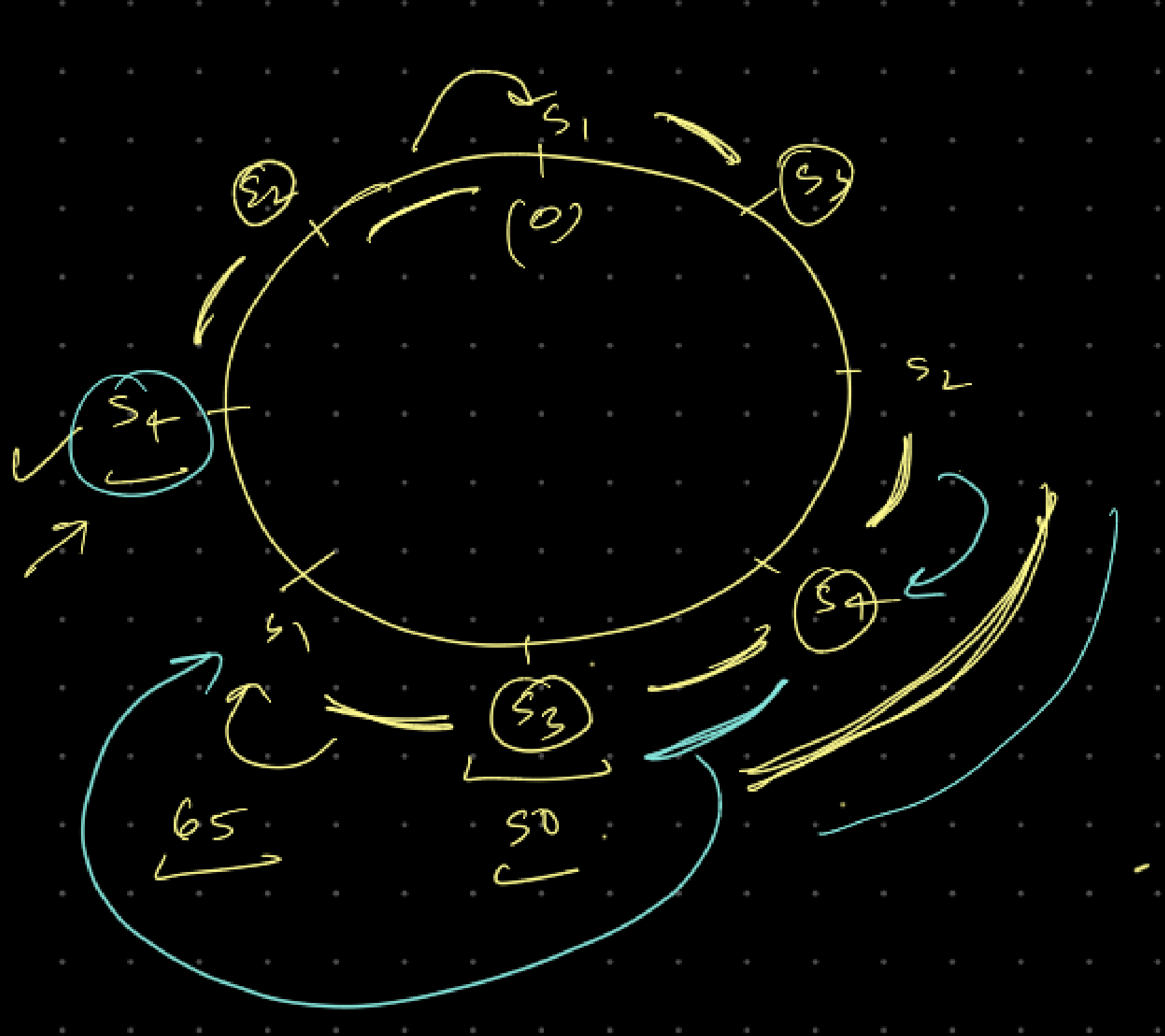
→ req → h(req)

h % 100 → 0 → 99

h % 100 → 0 → 25
0 → 25

% 4 → 0 → 3
S1 S2 S3 S4

Removing a server → load on a particular server spiked



S1 → 0
S2 → 25
S3 → 50
S4 → 75

0 → 100 [0, 15, 75]

J.W.T

↳ token + J.SON + web App

① Json . web Token → stateless ①

②. J.W.T hide data ② Yes (NO)

user ⇒ id + password → server → D.B

J.W.T token

subsequent R
J.W.T

client → J.W.T

stateless

J.W.T

api J.W.T → server → (X)

JWT → [header + payload] signature

server ⇒ (h(h) + h(p)) encrypts key ⇒ sign

3rd part of token

{Refresh Tokens
J.W.T}

J.W.T → 15 mins

userid
password → server

R.T
J.W.T

15 mins

client → R.T → server

J.W.T
R.T

R.T

R1 → ~~use~~
R2 → ~~used~~
R3 → use

Login → R1, J1

R1 → R2 J2

R2, J2 → R2 → R3 T3

R2

R3 T3